

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
Second Semester of B. Pharm. Examination
University Theory Examination April-May 2016
PH120 Pharmaceutical Engineering-II

Date: 10/05/2016, Tuesday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions: There are three sections in this question paper.

- 1. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.**
- 2. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.**
- 3. Figures to right indicate marks.**
- 4. Draw neat sketches wherever necessary.**

Q 1 Attempt all questions. Each question is of one mark. 20

- 1. Name the process by which water passes directly from the solid phase to the vapor phase**
 - a) Sublimation
 - b) Crystallization
 - c) Distillation
 - d) Evaporation
- 2. What is the source of heat in most of the Evaporators?**
 - a) Steam
 - b) Oil bath
 - c) Hot water
 - d) Coal
- 3. Normally evaporation process is carried out at one of the following experimental conditions**
 - a) Above the boiling temperature
 - b) Below the boiling temperature
 - c) At the boiling points
 - d) Room temperature
- 4. Separation of liquid by distillation is based on one of the following principles?**
 - a) Boiling point
 - b) Vapour pressure
 - c) Miscibility
 - d) Viscosity
- 5. Deviations from the Raoult's law observed in _____ solutions**
 - a) Ideal
 - b) Perfect
 - c) Real
 - d) None
- 6. Constant boiling solutions are also known as**
 - a) Zeotropic solutions
 - b) Zootropic solution
 - c) Azeotropic solutions
 - d) None

7. _____ type of distillation technique is used for extracting most of the volatile oils such as clove , eucalyptus
- Flash distillation
 - Azeotropic distillation
 - Simple distillation
 - Steam distillation
8. Which part in the distillation apparatus represents heat exchanger?
- Receiver
 - Still
 - Adopter
 - Condenser
9. Which of the following dryer is known as Lyophilizer?
- Freeze dryer
 - Fluidised dryer
 - Spray dryer
 - Vacuum dryer
10. In drying process, the final product is in the form of
- Solid
 - Slurry
 - Solution
 - Concentrate
11. Which condition is highly critical for drying operation?
- Temperature
 - Moisture
 - Pressure
 - Volume
12. Which one of the following types of product is having equilibrium moisture content of practically Zero?
- Non porous and soluble
 - Non porous and insoluble
 - Porous and insoluble
 - Porous and soluble
13. Increase the surface area, filtration will _____.
- Decrease
 - Constant
 - Increase
 - None effect
14. For sterile filtration which process is use
- Drum filtration
 - Leaf filtration
 - Membrane filtration
 - None of the above
15. Which instrument can work on principal of filtration centrifugation?
- Cooling Centrifuge
 - High Speed Centrifuge
 - Evaporation centrifuge
 - Basket Centrifuge
16. What is the principal of conical disc centrifuge?
- Filtration
 - Evaporation
 - Both (a) & (b)
 - Sedimentation

- D Explain principle, working and advantages with labeled diagram of steam distillation. 05
- E Enlist the methods for measurement of humidity. Briefly describe wet bulb and dry bulb method. 05
- F Write a note on filter aids. 05
- Q 4 Attempt any FOUR of the following.
- A Write the construction and working of perforated basket centrifuge with labeled diagram. 05
- B (1) If a mixture of air and water vapour at 80° F and 760 mm Hg pressure has humidity of 0.010 lb water vap/lb dry air, and if it saturated with water vapour at the same temperature and pressure the humidity is 0.0233 lb water vap/lb dry air. What will be the % humidity (H_p). 02
- (2) Give the salient features of Psychrometric chart. 03
- C Describe the various mechanisms of drying process. 05
- D Give construction, working and applications of sedimentation type centrifuge in pharmacy. 05
- E Classify the dryer as per its mechanism. Briefly describe the terms Equilibrium Moisture content (EMC) and Free Moisture Content (FMC). 05
- F Describe the construction and working of a fluidized bed dryer. 05
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